Power Electronics Laboratory Manual

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

SCHOOL OF ENGINEERING AND SCIENCES (SEAS)

SRM UNIVERSITY-AP

Mangalagiri, Neerukonda,
Andhra Pradesh 522502

DO'S AND DON'TS IN THE LAB

DO'S: -

1. Students should carry observation notes and records completed in all aspects.
2. Correct specifications of the equipment have to be mentioned in the circuit diagram.
3. Students should be aware of the operation of equipment's.
4. Students should take care of the laboratory equipment's/ Instruments.
5. After completing the connections, students should get the circuits verified by the Lab Instructor.
6. The readings/waveforms must be shown to the concerned faculty for verification.
7. Students should ensure that all switches are in the OFF position to remove the connections before leaving the laboratory.
8. All patch cords and chairs should be placed properly in their respective positions.
9. The steps for simulation of different tools should be properly known to the students for the software related laboratory.

DON'Ts: -

1. Come late to the Lab.
2. Make or remove the connections with power ON.
3. Switch ON the power supply without verification by the instructor.
4. Switch OFF the machine with load.
5. Leave the lab without the permission of the concerned faculty.

LIST OF EXPERIMENTS

Experiment No: **Date:**

Study the I-V characteristics of Silicon Controlled Rectifier.

Aim:

1. To plot the static V-I characteristics of the given SCR.
2. To find the Latching and Holding currents of the given SCR.

Apparatus:

1. Device characteristics trainer kit
2. Digital Multimeter - 3 numbers
3. Connecting wires

Circuit diagram:

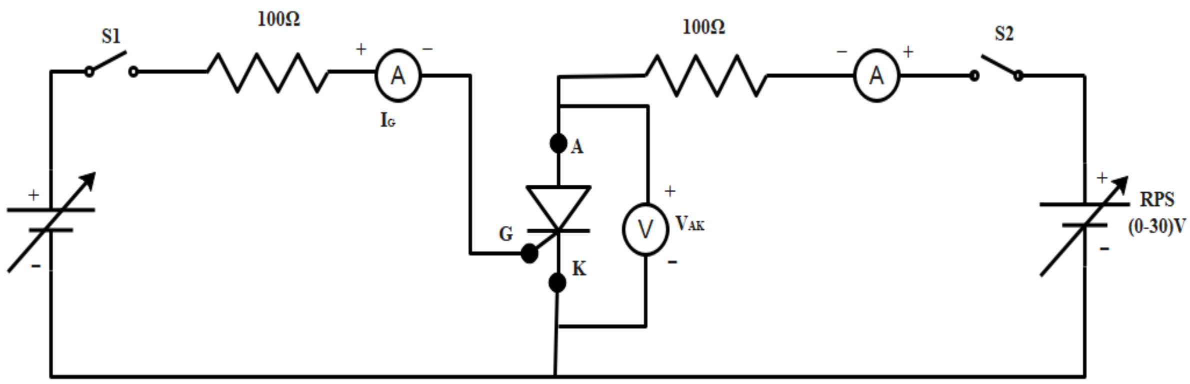


Fig. 1.1 Circuit diagram of SCR characteristics

Precautions:

1) Keep the switches, S_1 and S_2 in OFF position & the POT-1 and POT-2 in minimum position.

Connection procedure - for characteristics of SCR:

1) Connect SCR- P_1 terminal to P_{13} terminal in the front panel by using patch cords.
2) Connect SCR- P_2 terminal to P_{15} terminal in the front panel by using patch cords.
3) Connect SCR- G P_3 terminal to P_{12} terminal in the front panel by using patch cords.
4) Connect Power supply output V_1 & V_2 terminals to P_7 & P_8 terminals, respectively.
5) Connect Power supply output V_3 & V_4 terminals to P_{18} & P_{19} terminals, respectively.
6) Connect digital multimeter (in DC AMPS mode) positive & negative terminals to P_{14A} & P_{14} terminals, respectively.
7) Connect digital multimeter (in DC AMPS mode) positive & negative terminals to P_9 & P_{10} terminals, respectively.
8) Connect digital multimeter (in DC VOLT mode) positive & negative terminals to P_{14B} & P_{16} terminals, respectively.

Experimental Procedure:

1) Switch ON the trainer kit.
2) Switch ON toggle switch S_1 and set the value of gate current (I_G) to ___mA by using POT-1.

3) Switch ON toggle switch, S_2. Now vary the Anode-Cathode supply voltage gradually step by step by adjusting POT-2. Note down the corresponding values of V_{AK} & I_A.
4) Repeat the above procedure for another value of gate current.
5) The Point at which SCR fires gives the values of V_{Bo}.
6) Draw a graph of (V_{AK} & I_A) for different values of I_G.
7) Switch OFF the gate supply-Switch S_1.
8) Observe the ammeter (I_A) reading, by reducing the anode voltage by using POT-2. The point at which the ammeter reading suddenly goes to zero gives the value of Holding current (I_H).
9) Repeat the above procedure for another value of gate current.

Waveform:

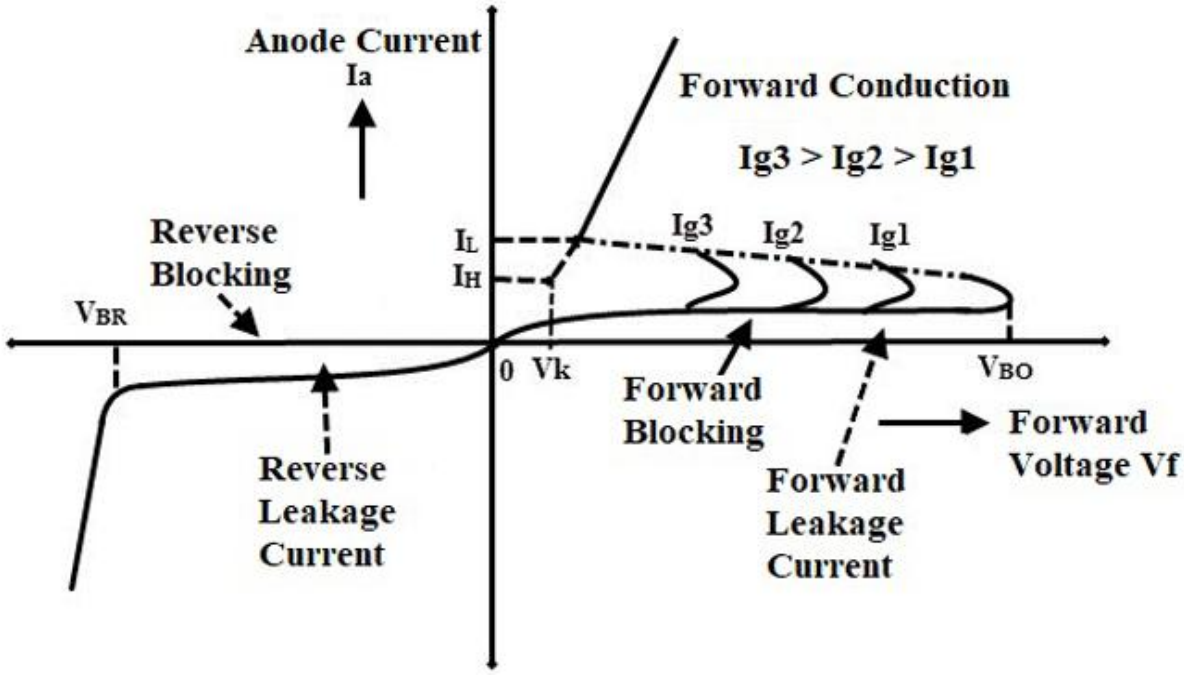


Fig. 1.2 Characteristics of SCR

Tabular column:

Gate Current I_{G1} = ________ mA

I_{G2} = ______ mA

S.No.	$V_{AK}(Volts)$	I_A(__ A)
1		
2		
3		

4		
5		
6		
7		
8		
9		
10		
S.No.	$V_{AK}(Volts)$	$I_A(\_\ \ A)$
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

Result:

Experiment No: **Date:**

Study the I-V characteristics of MOSFET

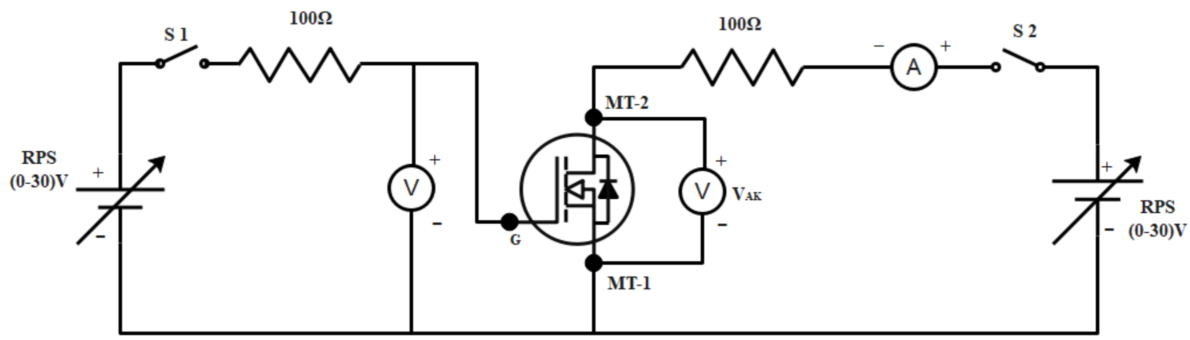
Aim:

To study the static characteristics of a MOSFET.

Apparatus:

1. Device characteristics trainer
2. Digital multimeter-3 number

Circuit Diagram:



Precaution:

1. Keep the S1 switch is on OFF position & keep the POT-1 is in minimum position
2. Keep the S2 switch is on OFF position & keep the POT-2 is in minimum position.

Connection Procedure for Characteristics Of MOSFET:

1. Connect MOSFET-P_{23} terminal to P_{33} terminal in front panel by using patch cords.
2. Connect MOSFET-P_{24} terminal to P_{35} terminal in front panel by using patch cords.
3. Connect MOSFET-G terminal (P_{25}) to P_{30} terminal in front panel by using patch cords.
4. Connect Power supply output terminals V_1 & V_2 to P_{26} & P_{27} terminals in the front panel by using patch cords.
5. Connect Power supply output terminals V_3 & V_4 to P_{38} & P_{39} terminals in the front panel by using patch cords.
6. Connect Digital Multimeter (in DC AMPS Mode) (+) Positive and (-) Negative terminals to P_{32} & P_{31} respectively.
7. Connect Digital Multimeter (in DC VOLT Mode) (+) Positive and (-) Negative terminals to P_{34} & P_{36} respectively.
8. Connect Digital Multimeter (in DC VOLT Mode) (+) Positive and (-) Negative terminals to P_{28} & P_{29} respectively.

Procedure For Drain Characteristics:

1. Switch ON the trainer.
2. Switch ON- SI Toggle switch and the value of Gate voltage (V_{GS}) is set to convenient value (say______) by using POT-1.

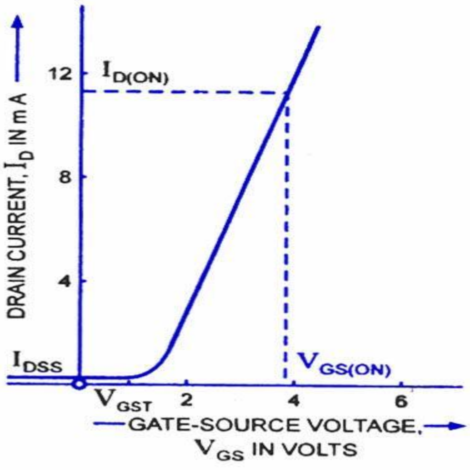
3. Switch ON-S2 Toggle switch, by varying the supply voltage gradually by Adjusting the POT-2 in variable DC source-2-Note down the corresponding value of V_{DS} & I_D table.
4. Repeat the above procedure with different gate source voltage (V_{GS}) and note down the value of V_{DS} & I_D in table.
5. Draw a graph of I_D vs V_{DS} for different values of V_{GS}.

Procedure For Transconductance (Transfer) Characteristics

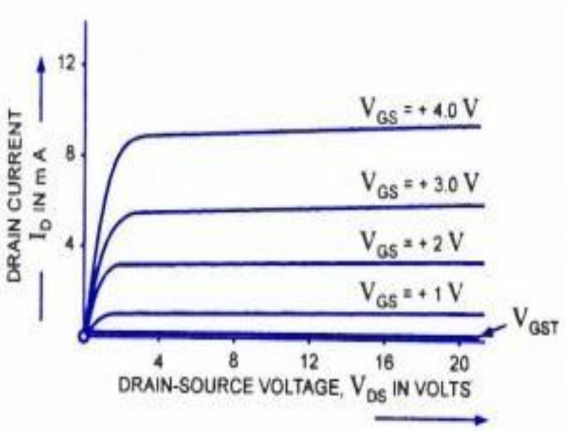
1. Switch ON the trainer.
2. Switch ON-S_2 Toggle switch, set V_{DS} at some voltage (_______V)
3. Switch ON-S_1 Toggle switch, by varying the supply voltage gradually by Adjusting the POT-1 in variable DC source-1, Note down the corresponding value of V_{GS} & I_D table.
4. Repeat the above procedure with different V_{DS} and note down the value of V_{GS} & I_D in table.
5. Draw a graph of I_D vs V_{GS} for different values.

Waveforms:

Transfer characteristics



Drain-Source characteristics



Transfer Characteristics:

S. No.	V_{DS1}= _____ (Volts)		V_{DS2}= _____ (Volts)	
	V_{GS} (V)	I_D (mA)	V_{GS} (V)	I_D (mA)
1				
2				
3				
4				
5				
6				

7				
8				
9				
10				

Drain Characteristics:

S. No.	V_{GS1}= ______ (Volts)		V_{GS2}= ______ (Volts)	
	V_{DS} (V)	I_D (mA)	V_{DS} (V)	I_D (mA)
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Threshold Voltage =

Result:

Experiment: **Date:**

Study The Performance of Single-Phase Half Controlled Rectifier

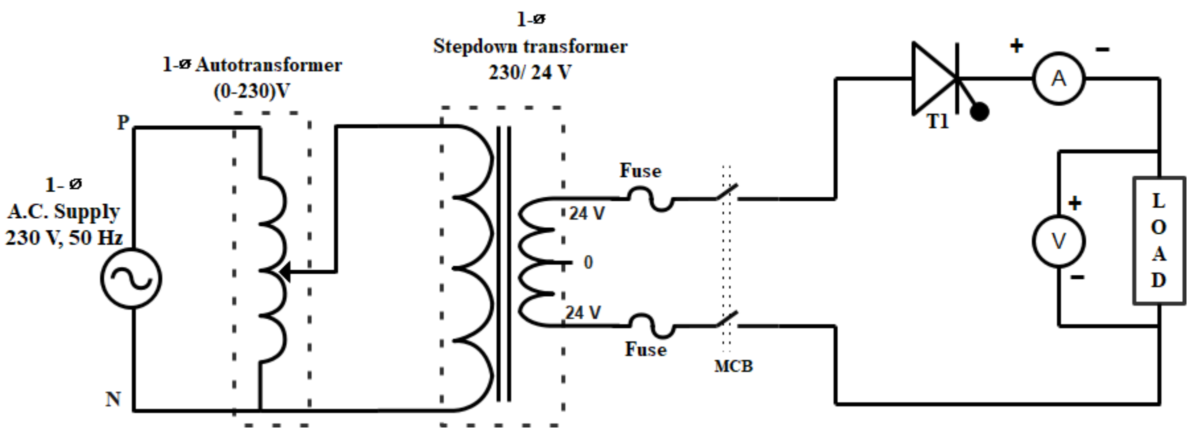
Aim:

To study the single-phase half wave converter with R load.

Apparatus required:

S. No	Components	Specifications	Quantity
1	Single phase SCR power module		1 No
2	Digital storage oscilloscope (DSO)		1 No
3	R, RL loads		1 No
4	Autotransformer	0-230 V	1 No
5	Step-down transformer	230 / 24V OR 48 V	1 No
6	Firing pulse controller		1 No
7	Connecting wires and probes		As per required

Circuit diagram:



Precautions:

1. While making the connections all the equipment kept in OFF position.
2. Check the PWM cable connection with controller & SCR power module.
3. Connect the ds pic programmer with firing pulse controller.

Procedure:

1. Connect single-phase input supply to SCR power module (back side).
2. Connect single-phase input supply to P_{IN} and N_{IN}, terminals in SCR power module through single-phase step-down transformer by using autotransformer (as per diagram).
3. Connect K_1-K_1, and G_1-G_1, by using patch cords.
4. Connect R load between K_1 & N_{IN} terminals in SCR power module.

5. Switch "ON" the power using the power ON/OFF switch in firing pulse controller unit.
6. Switch "ON" the power using the power ON/OFF switch in the SCR unit.
7. Switch "ON" single-phase supply to SCR unit.
8. Check the ZCD square pulse in test point connectors in the SCR power module front panel using CRO.
9. Switch on the power in firing pulse controller unit by using IRS switch.
10. Then switch ON the autotransformer, then apply 230v by using the autotransformer.
11. Press SW3 switch to increase the firing angle from 180-0 degrees.

Observation table:

R load: **R = 100Ω**

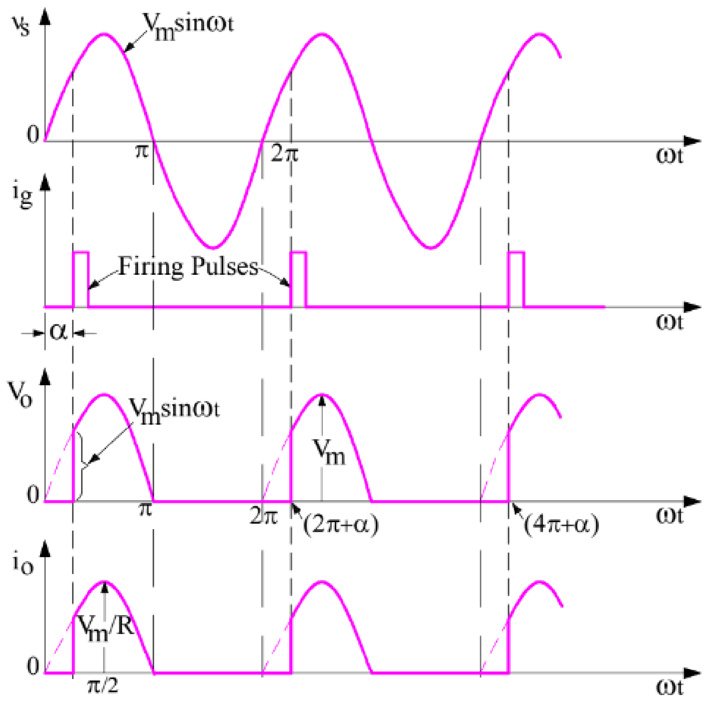
S. No	Firing angle (α) degree	Average Output Voltage (V_{oavg})		Average Output Current (I_{oavg})		Rms Output Voltage (V_{orms})		Rms Output Current (I_{rms})	
		Theoretical	Practical	Theoretical	Practical	Theoretical	Practical	Theoretical	Practical
1									
2									
3									
4									
5									
6									
7									

R-L load: R= L=

S. No	Firing angle (α) degree	Average Output Voltage (V_{oavg})		Average Output Current (I_{oavg})		Rms Output Voltage (V_{orms})		Rms Output Current (I_{rms})	
		Theoretical	Practical	Theoretical	Practical	Theoretical	Practical	Theoretical	Practical
1	180								
2	150								
3	120								
4	90								
5	60								

6	30								
7	0								

Output waveforms:
With R-load



Theoretical calculations:

$$V_{0avg} = \frac{V_m}{2\pi}(1 + cos\ \alpha) = \frac{\sqrt{2}\ x\ 48}{2\ \pi}\ (1 + cos\ 30^0)$$

$I_0 \quad = \frac{V_0}{R} =$

$$V_{rms} = \frac{Vm}{2\sqrt{\pi}}\left[(\pi - \alpha) + \left(\frac{sin\ 2\ \alpha}{2}\right)\right]^{1/2}$$

At $\boldsymbol{\alpha} = 60^0$

$$V_{rms} = \frac{48\sqrt{2}}{2\sqrt{\pi}}\left[\left(\pi - \ 60\ x\,\frac{\pi}{180}\right) + \frac{(sin2\ (60^0))}{2}\right]^{1/2}$$

Result:

Experiment: **Date:**

Study The Performance of Single Phase Fully Controlled Bridge Converter.

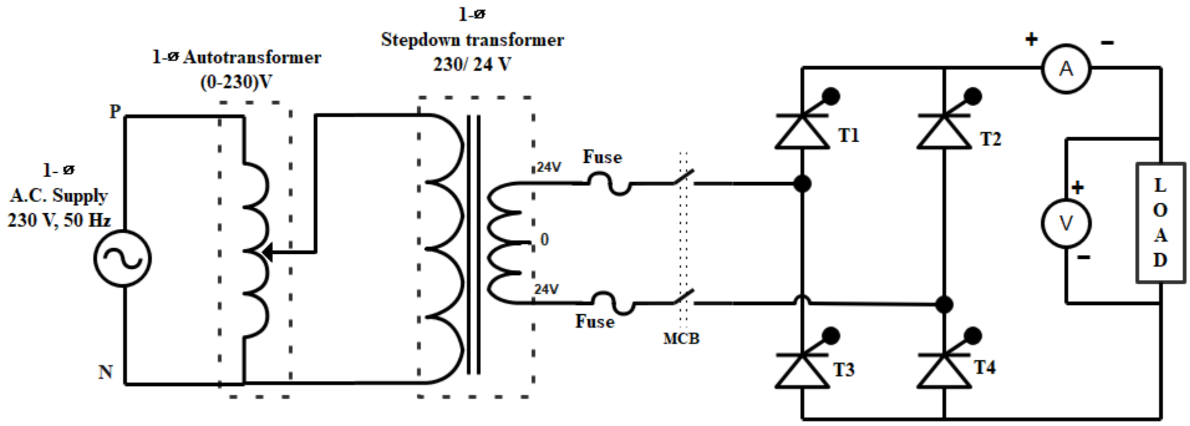
Aim:

To study the single phase fully controlled full bridge converter with R & RL load

Apparatus required:

S. No	Components	Specifications	Quantity
1	Single phase SCR power module		1 No
2	Digital storage oscilloscope (DSO)		1 No
3	R, RL loads	100 Ω, 100mH	1 No
4	Autotransformer	0-230 V	1 No
5	Step-down transformer	230 V / 24V	1 No
6	Firing pulse controller		1 No
7	Connecting wires and probes		As per required

Circuit diagram:



Precautions:

1. Keep the input MCB in off condition in the SCR power module.
2. Keep the power ON/OFF switch is in off position in PLVSI-01.
3. Connect 230 V_{AC} input supply to P, N terminals (through isolation & auto transformer).
4. Check the PWM cable connection with controller & SCR power module.
5. Connect the ds pic programmer with microcontroller

Procedure:

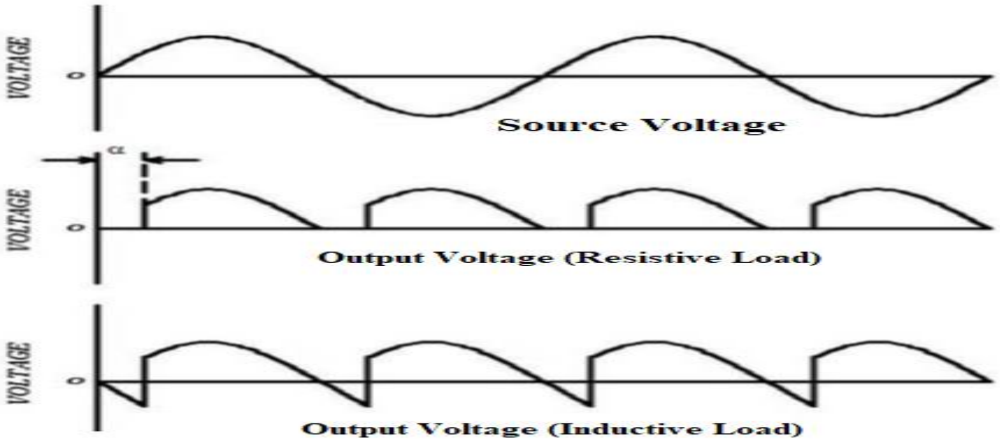
R-load

1. Connect single-phase input supply to SCR power module (back side).
2. Connect single-phase input supply to P_{IN} and N_{IN} terminals in SCR power module through single-phase autotransformer (as per diagram).
3. Connect firing pulses for each SCR from the firing pulse controller (ex: G_1-G_1, K_1-K_1).
4. Connect K_1 to K_3 (all TWO to be shorted using patch chords or 5A wires).
5. Connect A_2 to A_4 (all TWO to be shorted using patch chords or 5A wires).
6. Connect R-load between K_3 & A_4 terminals in SCR power module.
7. Switch "ON" the power using the power ON/OFF switch in the firing pulse controller unit.
8. Switch "ON" the power using the power ON/OFF switch in the SCR unit.
9. Switch "ON" single-phase supply to SCR unit.
10. Check the ZCD square pulse in test point connectors in the SCR power module front panel using CRO.
11. Switch on the power in firing pulse controller unit by using IRS switch in firing pulse controller unit, the display will be ON, then select open-loop function by using INC key.
12. Then switch ON the autotransformer, then apply 230v by using the autotransformer.
13. Press SW3 switch to increase the firing angle from 180-0 degrees.

RL load:

Repeat the above procedure by replacing R load with RL load.

Voltage waveforms for R and RL load:



R load: R=

S. No	Firing angle (α) degree	Average Output Voltage (V_{oavg})		Average Output Current (I_{oavg}) mA		Rms Output Voltage (V_{orms})		Rms Output Current (I_{rms})	
		Theoretical	Practical	Theoretical	Practical	Theoretical	Practical	Theoretical	Practical
1	180								
2	150								
3	120								
4	90								
5	60								
6	30								
7	0								

R-L load: R= L=

S. No	Firing angle (α) degree	Average Output Voltage (V_{oavg})		Average Output Current (I_{oavg})		Rms Output Voltage (V_{orms})		Rms Output Current (I_{rms})	
		Theoretical	Practical	Theoretical	Practical	Theoretical	Practical	Theoretical	Practical
1	180								
2	150								
3	120								
4	90								
5	60								
6	30								
7	0								

Theoretical calculations:

R load:

$$V_o = \frac{2V_m}{\pi} cos\alpha \qquad I_{avg} = \frac{V_{avg}}{R}$$

Observations:

Result:

Experiment: **Date:**

Study the performance of single-phase semi-controlled bridge converter.

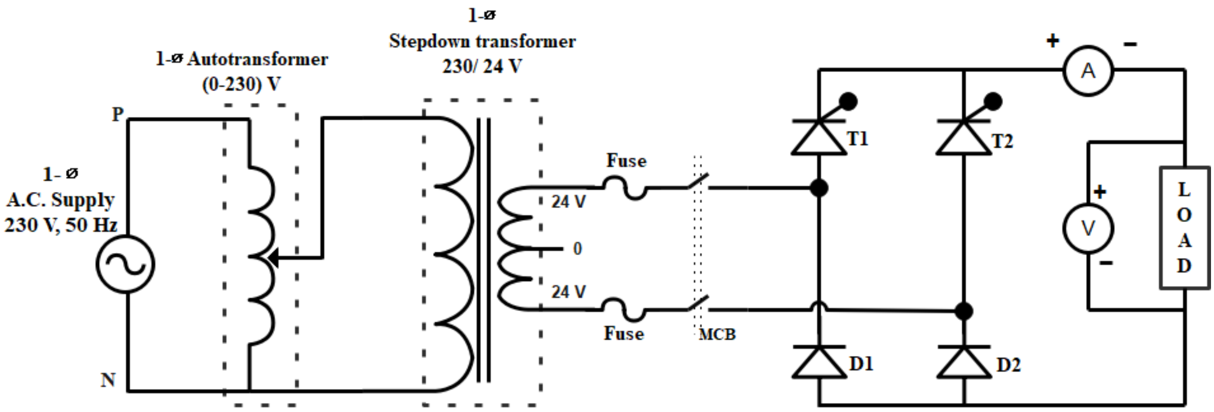
Aim:

To study the performance of single-phase semi-controlled rectifier.

Apparatus required:

S. No	Components	Specifications	Quantity
1	Single phase SCR power module		1 No
2	Digital storage oscilloscope (DSO)		1 No
3	R, l loads	R = 100 Ω & L = 100mH	1 No
4	Autotransformer	0-230 V	1 No
5	Step-down transformer	230 / 24V	1 No
6	Firing pulse controller		1 No
7	Connecting wires and probes		As per required

Circuit diagram:



Precautions:

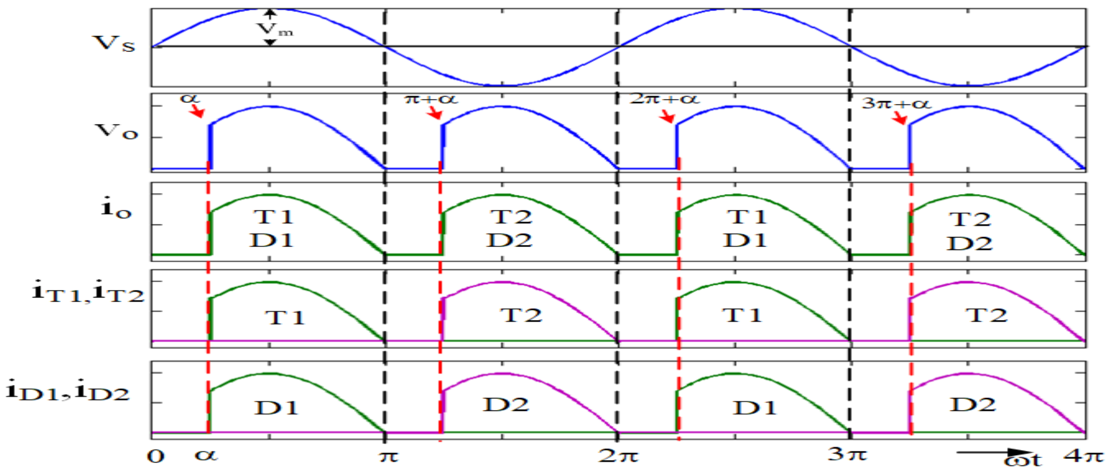
1. Keep the input MCB is in off condition in SCR power module
2. Keep the power ON/OFF switch is in off position in PLVSI-01.
3. Connect 230V AC input supply to P, N terminals (through isolation & auto transformer).
4. Check the PWM cable connection with controller & SCR power module.
5. Connect the ds pic programmer with firing pulse controller.

Procedure:

1. Connect single-phase input supply from step-down transformer to P_{IN}, N_{IN} terminals.
2. Connect gating signal for T_1 and T_3 from G_1-G_1, K_1-K_1, G_3-G_3 and K_3-K_3 respectively using patch chords.
3. Connect the load between the terminals K_3 & AD_2 terminals in SCR power module
4. Switch "ON" the power using power ON/OFF switch in firing pulse controller unit.
5. Switch "ON" the power using power ON/OFF switch in SCR unit.
6. Check the ZCD square pulse in test point connectors in SCR power module front panel using CRO.
7. Switch on the power to firing pulse controller unit by using IRS switch
8. Then switch ON main supply and apply 230 V by using the autotransformer.
9. Press SW3 switch to increase the firing angle from 180-0 degree

R load:

Expected waveforms:



Observation table:

R=100 ohm

S. No	Firing angle (α) degree	Average output voltage (V_{oavg})		Average output current (I_{oavg})		Rms output voltage (V_{orms})		Rms output current (I_{rms})	
		Theoretical	Practical	Theoretical	Practical	Theoretical	Practical	Theoretical	Practical
1	0								
2	30								
3	60								
4	90								

5	120								
6	150								
7	180								

Theoretical calculations:

$$V_{0avg} = \frac{V_m}{\pi}(1 + cos\alpha)$$

$$V_{orms} = \frac{V_m}{\sqrt{2\pi}}\left[(\pi - \alpha) + \frac{1}{2}sin(2\alpha)\right]^{\frac{1}{2}}$$

Results:

Experiment: **Date:**

Study the performance of the DC-DC step-down (BUCK) chopper.

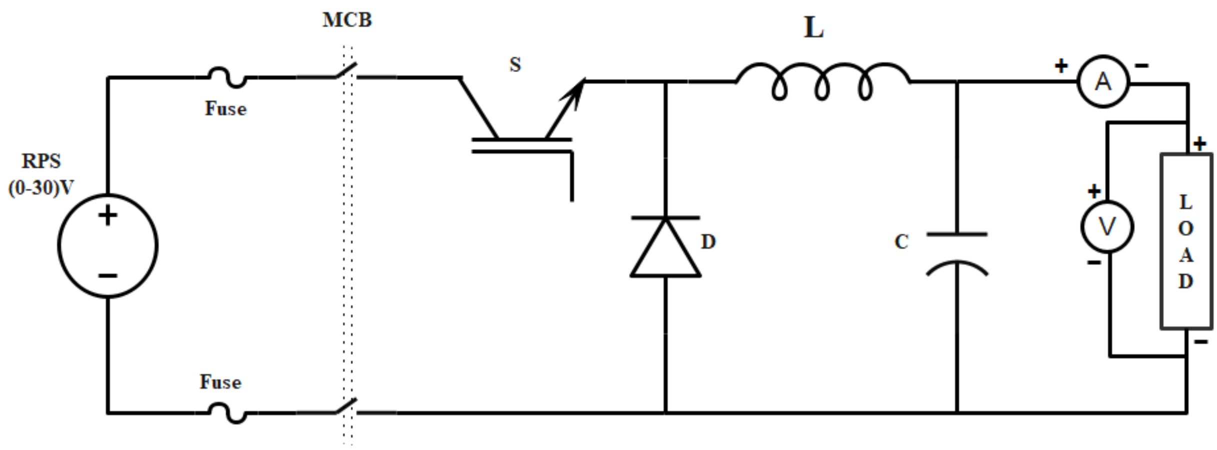
Aim:

To study the operation of a dc-dc buck converter.

Apparatus required:

S. No	Components	Range	Quantity
1	DC-DC buck converter kit		1 No
2	Digital storage oscilloscope (DSO)		1 No
3	R load	100 Ω	1 No
4	Regulated power supply (RPS)	0-30 V	1 No
5	Connecting wires and probes		As per required

Circuit diagram:



Precautions:

1. Keep the power ON/OFF switch is in off position.
2. Keep the ***sw₁*** switch in OL position in DC-DC buck converter unit.
3. Keep DC MCB-1 in off position.

Procedure:

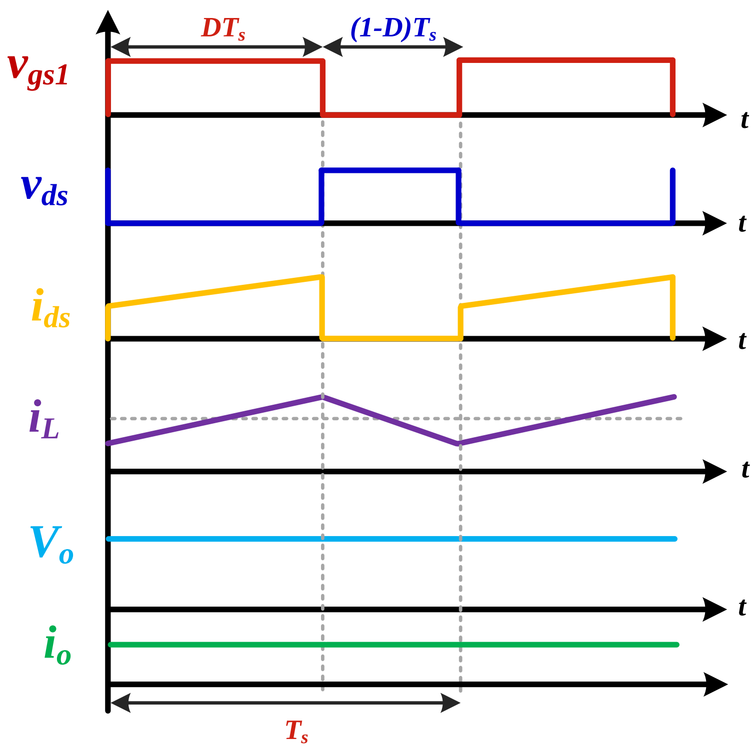
1. Connect the circuit as per circuit diagram.
2. Set the switching frequency at 10 kHz and connect the load resistance.
3. Adjust the duty cycle from 10% to 90% in steps at 10 and at each duty cycle measure input current, output current and output voltage.
4. Tabulate the results and compare the output voltage with theoretical voltage and calculate error.
5. Set the frequency to 20 kHz and repeat the same procedure.

Observation table:

Switching frequency (f_s)=10KHz, **Input voltage (V_{dc})** = 30V, **Load resistor (R)** = 100**Ω.**

S. No	Duty Cycle (D)	Input Current (I_{dc})	Output Voltage (V_0) (Practical)	Output Voltage (V_0) (Theoretical)	$I_{o(mA)}$	$P_{dc} = V_{dc}*I_{dc}$	$P_o = V_o*I_o$	η %
1	10							
2	20							
3	30							
4	40							
5	50							
6	60							
7	70							
8	80							

Expected waveforms:



Theoretical calculations:

$D = \frac{V_0}{V_S} \Longrightarrow V_0 = D\,V_S$

$P_0 = V_{dc} \times I_{dc}$

$P_0 = V_0 \times I_0$

$\boldsymbol{\eta} = \frac{P_0}{P_{dc}} =$

Result:

Experiment: **Date:**

Study the performance of DC-DC step-up (BOOST) chopper

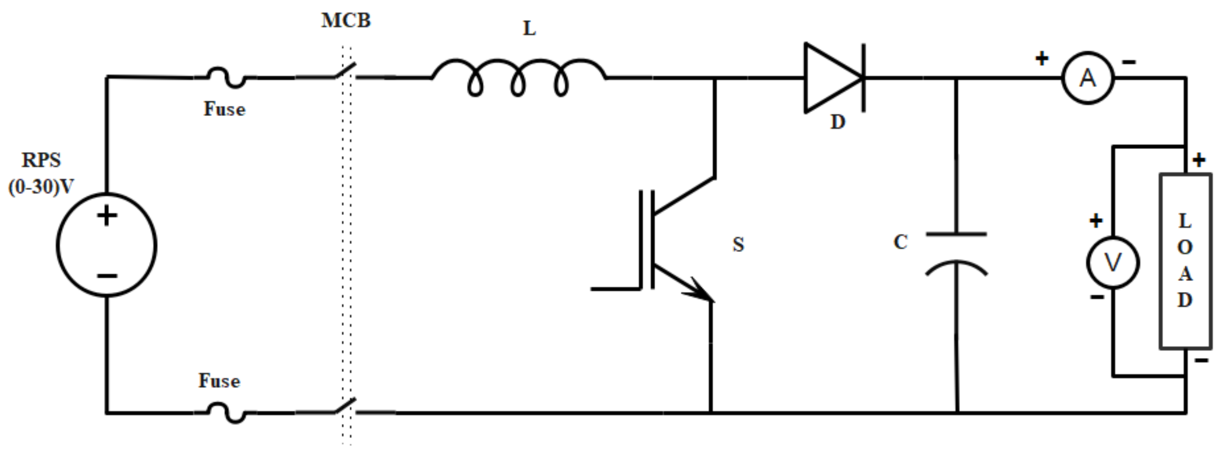
Aim:

To study the working of DC-DC boost converter.

Apparatus required:

S. No	Components	Specifications	Quantity
1	DC-DC boost converter kit		1 No
2	Digital storage oscilloscope (DSO)		1 No
3	R Load	100 Ω	1 No
4	Regulated power supply (RPS)	0-30 V	1 No
5	Connecting wires		As per required

Circuit diagram:



Precautions:

1. Keep the Power ON/OFF switch is in off position.
2. Keep the sw_1 switch is in OL position in DC-DC boost converter unit.
3. Keep DC MCB-1 in off position.

Procedure:

1. Connect to the circuit as shown in circuit diagram.
2. Set the switching frequency at 10 kHz and connect the load.
3. Adjust the duty cycle from 10% to 90% in steps at 10 and at each duty cycle measure input current, output current and output voltage.
4. Tabulate the results and compare the output voltage with theoretical voltage and calculate error.

5. Set the frequency to 20 kHz and repeat the same procedure.

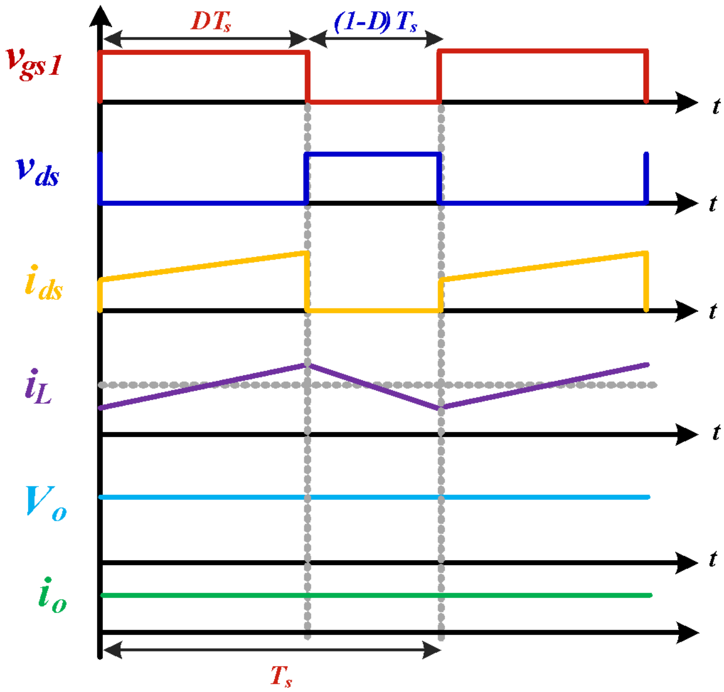
Observation table:

Switching Frequency (f_s)=10KHz **Input Voltage (V_{dc})** = 24V **Load Resistor (R)** = 100**Ω**

S. No	*Duty Cycle (D)*	Input Current (I_{dc})	V_o *(Practical)*	V_o *(theoretical)*	I_o	$P_{dc} = V_{dc} * I_{dc}$	$Po = V_o * I_o$	η
1	10							
2	20							
3	30							
4	40							
5	50							
6	60							

Expected waveforms:

Theoretical calculations:



$$V_0 = \frac{1}{1-D} V_s$$

Duty cycle at 30 %

V_0 =

P_{dc} = V_{dc} x I_{dc} =

P_0 = V_0 x I_0 =

η = $\frac{P_0}{P_{dc}}$ =

Result:

Experiment No: **Date:**

Study the performance of single-phase AC voltage controller.

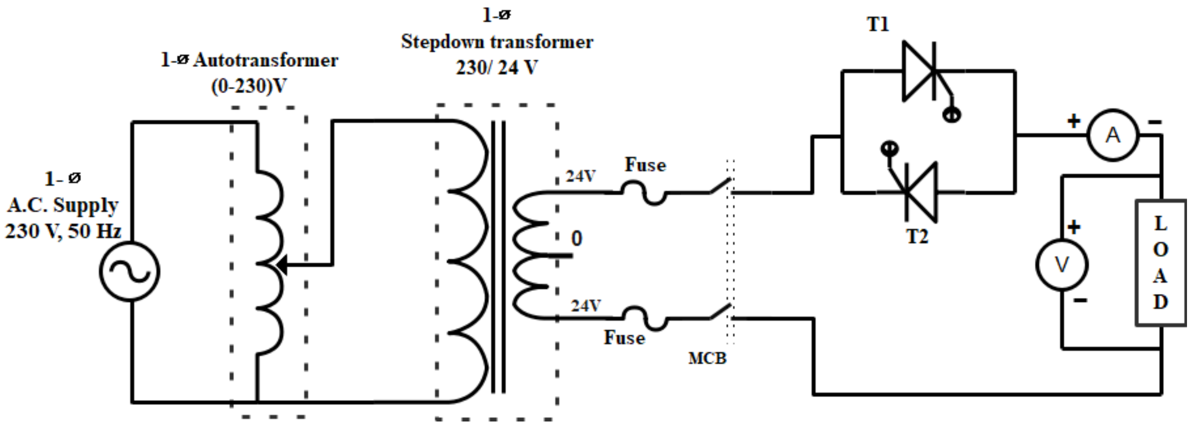
Aim:

To study the phase AC voltage regulator with R load.

Apparatus:

S. No	Components	Specifications	Quantity
1	Three Phase SCR Power module		1 No
2	CRO, CRO probes		1 No
3	R, Load		1 No
4	Step-Down Transformer	230 / 24V OR 48 V	1 No
5	Firing Pulse Controller		1 No
6	Connecting Wires		As per required

Circuit diagram:



Precautions:

1. Keep the input 1-phase MCB is in off condition in SCR unit.
2. Keep the input 1-phase autotransformer is in minimum condition.
3. Keep the power ON/OFF switch is in off position in SCR unit.
4. Keep the power ON/OFF switch is in off position in microcontroller unit.

Procedure:

1. Connect single-phase input supply to SCR power module (back side).
2. Connect single-phase input supply to P_{IN} N_{IN}, terminals in SCR power module through single-phase autotransformer with the help of step-down transformer. (As per diagram).

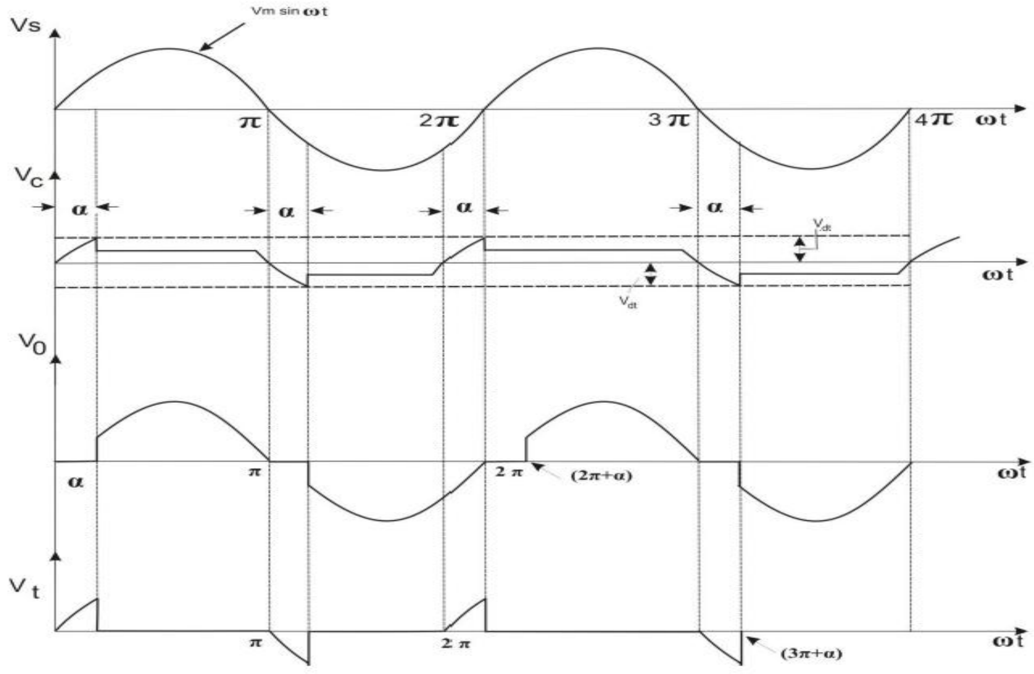
3. Connect K_1-A_2, by using patch chords. (5A wire)
4. Connect R-load between K_1 & N_{in} terminals in SCR power module.
5. Switch "ON" the power using power ON/OFF switch in firing pulse controller unit.
6. Switch "ON" the power using power ON/OFF switch in SCR unit.
7. Switch "ON" single-phase supply to SCR unit.
8. Check the ZCD square pulse in test point connectors in SCR power module front panel using CRO.
9. Switch on the power in firing pulse controller unit by using IRS switch.
10. Then switch ON the autotransformer, then apply 230V by using the autotransformer.
11. Press SW3 switch to increase the firing angle from 180-0 degree. Note down the corresponding values and observe the waveform.

Observation Table:

R load: R=

S. No	Firing angle (α) degree	Rms Output Voltage (V_{orms})		Rms Output Current (I_{rms})	
		Theoretical	Practical	Theoretical	Practical
1	180				
2	150				
3	120				
4	90				
5	60				
6	30				
7	0				

Expected waveforms:



Theoretical calculations:

$\mathbf{V_{rms}} = \boldsymbol{V_m}\sqrt{\frac{\mathbf{1}}{\mathbf{2}\boldsymbol{\pi}}\left\{(\boldsymbol{\pi}-\boldsymbol{\alpha})+\frac{\boldsymbol{sin2\alpha}}{\mathbf{2}}\right\}}$

If α = 120^0

degree to radians conversion

120 x $\frac{\pi}{180} = \frac{2\pi}{3}$ = 0.666667

= 48 $\sqrt{2}$ x $\sqrt{\frac{1}{2\pi}\left\{(3.1416-2.0944)+\frac{sin2(120)}{2}\right\}}$

=

V_{rms} = (theoretical).

Result:

Experiment No: **Date:**

Study the performance of single-phase step-down Cyclo Converter

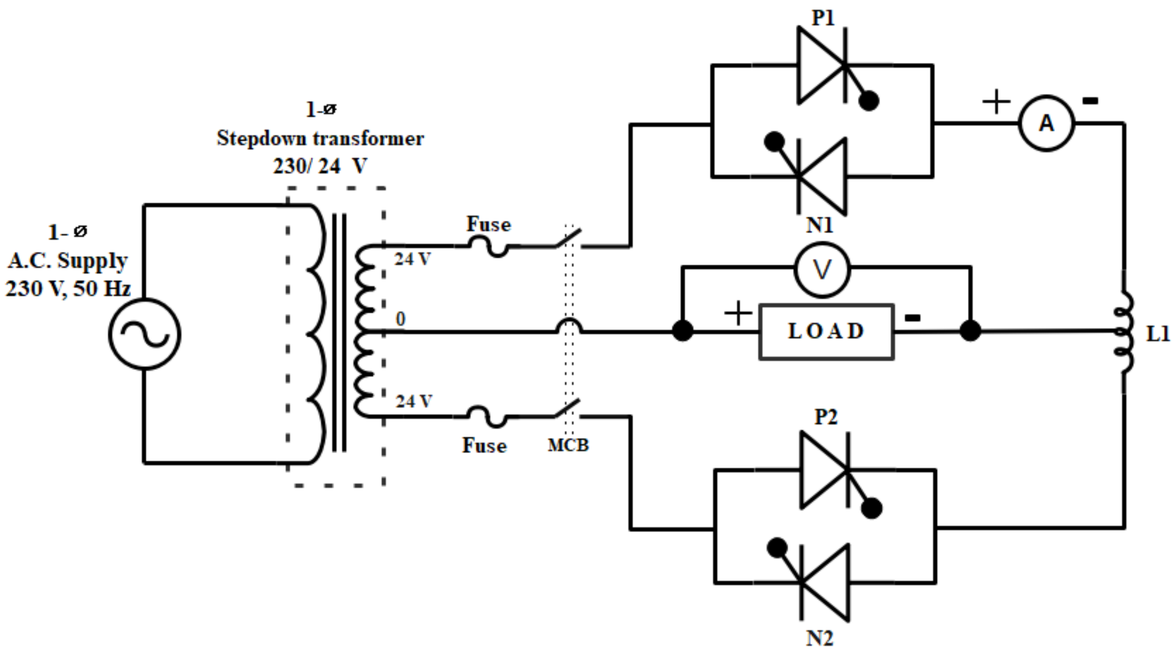
Aim:

To obtain the output waveforms of a single-phase cyclo-converter.

Apparatus:

S. No.	Component	Quantity
1	Cyclo Converter trainer kit	1
2	Digital storage oscilloscope (DSO)	1
3	Patch chords and probes	As per required
4	Firing Pulse Controller	1

Circuit diagram:



Precautions:

1. Keep the power ON/OFF switch is in off position in PLVSI-01.
2. Keep the power ON/OFF switch is in off position in cyclo converter trainer kit.

Procedure:

1. Connect D_1 terminal to D_2 terminal using patch chord.
2. Connect C_1 terminal to C_{1A} terminal using patch chord.
3. Connect C_3 terminal to C_{3A} terminal using patch chord.
4. Connect load R_1 terminal to C_2 terminal using patch chord.
5. Connect load R_2 terminal to C_4 terminal using patch chord.

6. Connect pulse cable from controller to SCR circuit using cable switch ON the power using power ON/OFF switch.
7. In LCD display 1:1, 1:2, 1:3, 1:4. 1:5, 1:6 mode will display.
8. Select any one mode (1:1) by using (Curser key) DEC Key.
9. Press ENTER key for selection, now 1:1 mode is selected.
10. Observe the wave form T_1, T_2, T_3 and T_4.
11. For Firing angel adjustment (for output voltage variation), press INC/DEC Key observe the phase

    angle variation in the available output voltage wave.
12. Observe and note down the load wave across R_1 & R_2 using DSO.

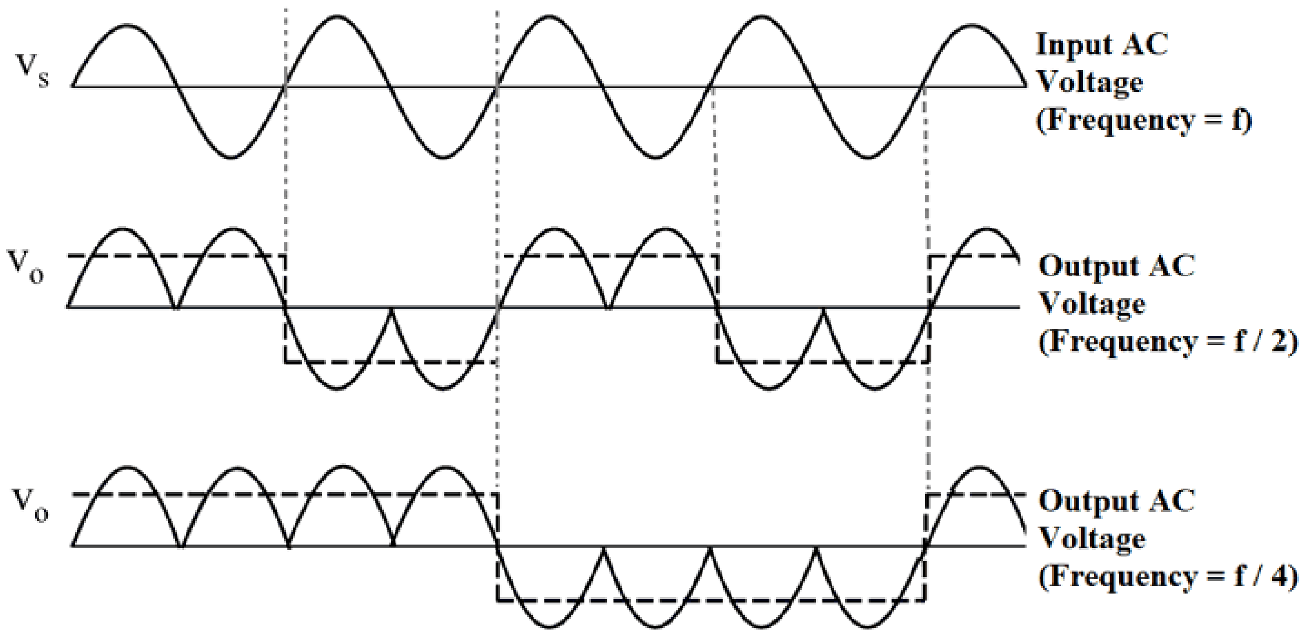
Tabular column:

S. No.	$\boldsymbol{f_s/1}$			$\boldsymbol{f_s/2}$			$\boldsymbol{f_s/3}$		
	Firing angle, α	Output voltage, V_0	Output current, I_0	Firing angle, α	Output voltage, V_0	Output current, I_0	Firing angle, α	Output voltage, V_0	Output current, I_0
1									
2									
3									
4									
5									

S. No.	$\boldsymbol{f_s/4}$			$\boldsymbol{f_s/5}$			$\boldsymbol{f_s/6}$		
	Firing angle, α	Output voltage, V_0	Output current, I_0	Firing angle, α	Output voltage, V_0	Output current, I_0	Firing angle, α	Output voltage, V_0	Output current, I_0
1									
2									
3									
4									
5									

Calculations:

Expected waveforms:



Result:

Experiment No: **Date:**

Study the 1-Ph full bridge Inverter with unipolar and bi-polar SPWM

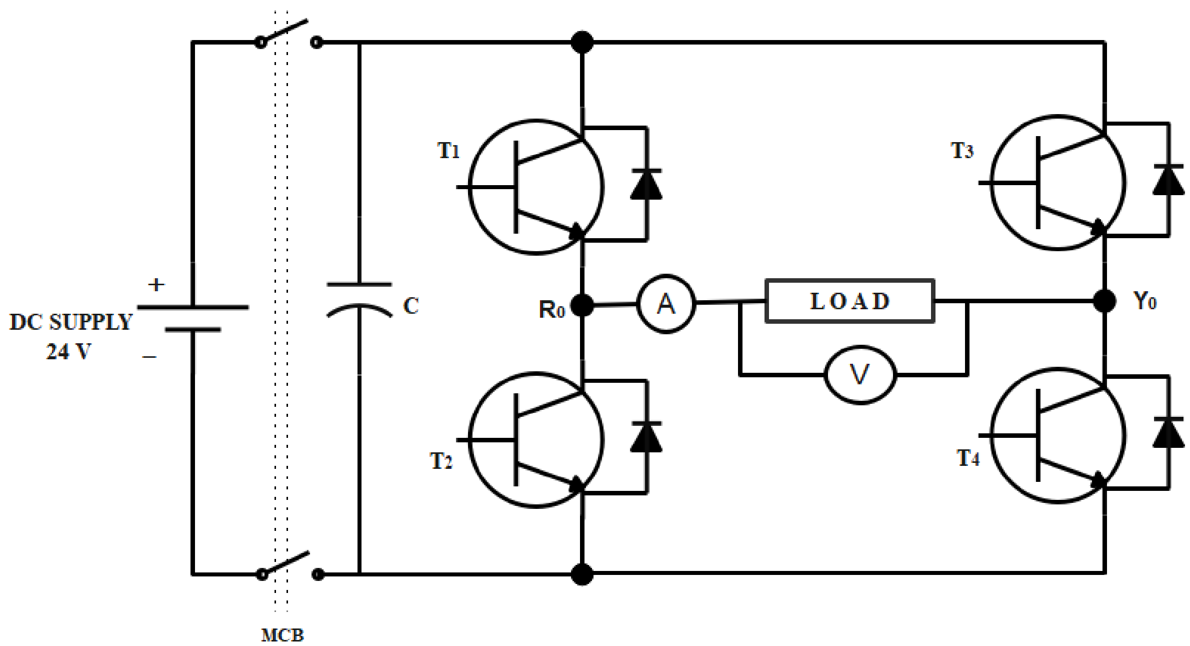
Aim:

Study of Uni-polar and bi-polar PWM based single-phase inverter

Apparatus required

S. No	Components	Specifications	Quantity
1	PLVSI-01 IGBT Power module		1 No
2	Digital storage oscilloscope (DSO)		1 No
3	Resistive load	100Ω	1 No
4	Regulated power supply (RPS)	0-30 V, 2A	1 No
5	Connecting wires and probes		As per required

Circuit Diagram:



Precautions

1. While doing the connections all equipment should be in OFF position.
2. Avoid loose connections
3. Always kept the current knob at maximum position in regulated power supply.

Connection Procedure:

1. Connect single-phase input supply to IGBT power module (back side) using 3 pin power cards.
2. Connect positive (+) and negative terminals of regulated power supply to positive and negative terminals of IGBT power module respectively.
3. Connect resistive load between R_0 &Y_0 terminals in IGBT power module through an ammeter.
4. Connect a voltmeter across the load to measure the output voltage.
5. Connect 34 pin flat cable to microcontroller (P_A) and IGBT power module (P_B) (pulse input connector).
6. Switch "ON" the power using the power ON/OFF switch in the IGBT power module.
7. Switch "ON" the power using power ON/OFF switch in RPS then, apply 24v supply by using voltage knob.
8. The Microcontroller display will be like below (after displaying name & address)

   SINGLE-PHASE

   THREE-PHASE
9. Press INC for SINGLE PHASE SELECTION
10. The Microcontroller display will be like below

    SQUARE WAVE INVERTER
11. Press DEC for selection (Enter) The Microcontroller display will be like below
12. Set Frequency --- HZ
13. Press INC for frequency increment & DEC for frequency decrement (Note the frequency varied from - 0 to 50hz) Switch "ON" MCB in IGBT Power Module.
14. Check the pulse output Test point connectors in the IGBT power Module front Panel using DSO.
15. Ensure the availability of all 4 pulses using DSO.
16. Press INC switch to increase the FREQUENCY.
17. For every frequency note down output voltage across load & tabulate to reset the function, Press *RST* reset switch.

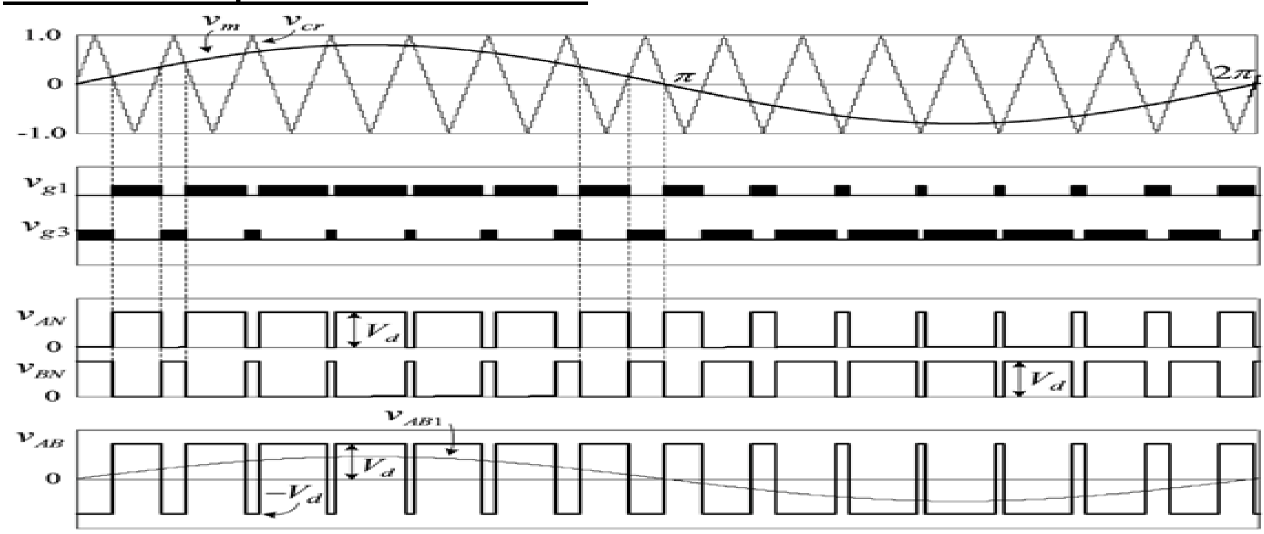
S. No.	Set Frequency (In Microcontroller)	Output Voltage V_0 (Theoretical)	Output Voltage V_0 (Practical)	Output Frequency f_0 (Theoretical)	Output Frequency f_0 (Practical)
1					
2					
3					
4					

Tabular column:

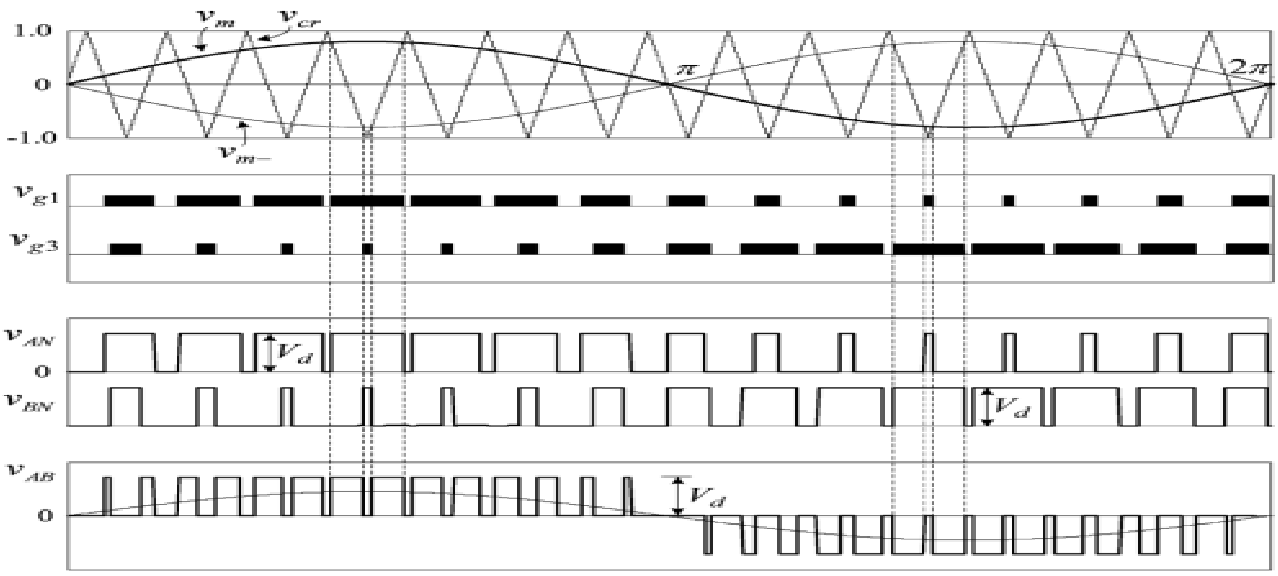
Calculations:

Expected waveforms:

<u>Waveforms of Bipolar Modulation Scheme</u>



<u>Waveforms of Unipolar Modulation Scheme</u>



Results:

Experiment No: **Date:**

Study the 3-Ph Inverter with 180- degree and 120-degree conduction mode.

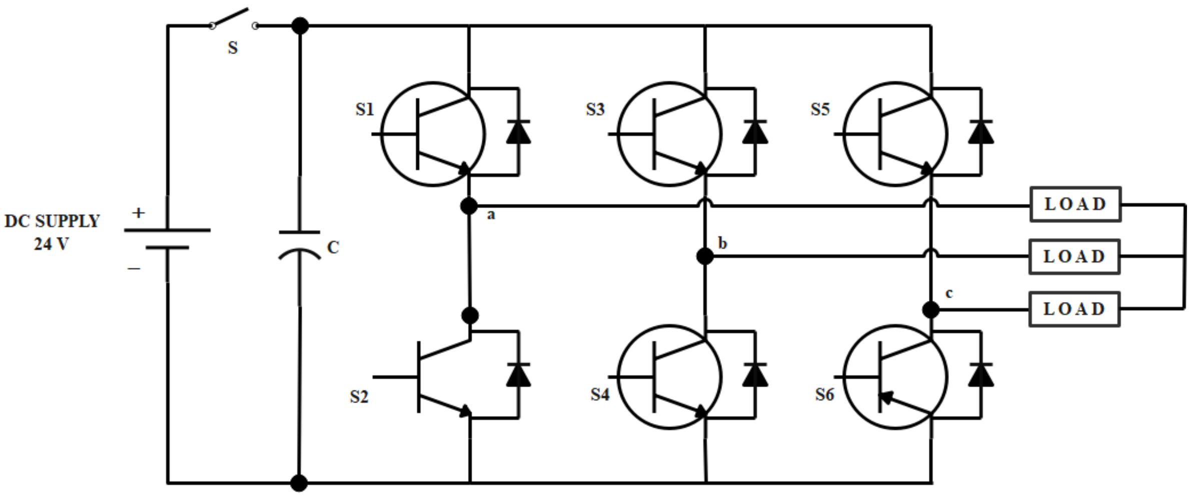
Aim:

To obtain three phase output wave forms for three phase PWM inverter

Apparatus Required:

S. No	Components	Specifications	Quantity
1	PLVSI-01 IGBT Power module		1 No
2	Digital storage oscilloscope (DSO)		1 No
3	Resistive load	100Ω	1 No
4	Regulated power supply (RPS)	0-30 V, 2A	1 No
5	Connecting wires and probes		As per required

Circuit Diagram:



Precautions:

1. Keep the Input 10 MCB is in off condition in PLVSI-01 unit.
2. Keep the Power ON/OFF switch is in off position in PLVSI-01.
3. Keep the Power ON/OFF switch is in off position in PLMC01-Microcontroller unit.
4. Connections are made as per the connection diagram.

Connection Procedure:

1. Connect 1 Phase input supply to IGBT power module (back side) using 3 pin power cards.
2. Connect 24V (or 48V) input supply to PN, terminals in IGBT power module through I phase Step down transformer (as per diagram).
3. Connect Rheostat -1 one terminal to R_0 terminals in IGBT power module.
4. Connect Rheostat -2 one terminal to Y_0 terminals in IGBT power module.
5. Connect Rheostat -3 one terminal to B_0 terminals in IGBT power module.
6. Connect All 3 Rheostats other terminals, it acts as neutral.
7. Connect 34 pin flat cable to microcontroller (P_A) and IGBT Power module (P_B) (pulse input connector).
8. Switch "ON" the power using power ON/OFF switch in 1GBT Power module.
9. Switch "ON" the power using power ON/OFF switch in Microcontroller unit.
10. The Microcontroller display will be like below (after displaying name & address)
    a. SINGLE PHASE
    b. THREE-PHASE
11. Press DEC for THREE PHASE SELECTION
12. The Microcontroller display will be like below

120 MODE INVERTER

180 MODE INVERTER

13. Press DEC for selection of 120-degree mode inverter
14. Press INC for enter
15. The Microcontroller display will be like below

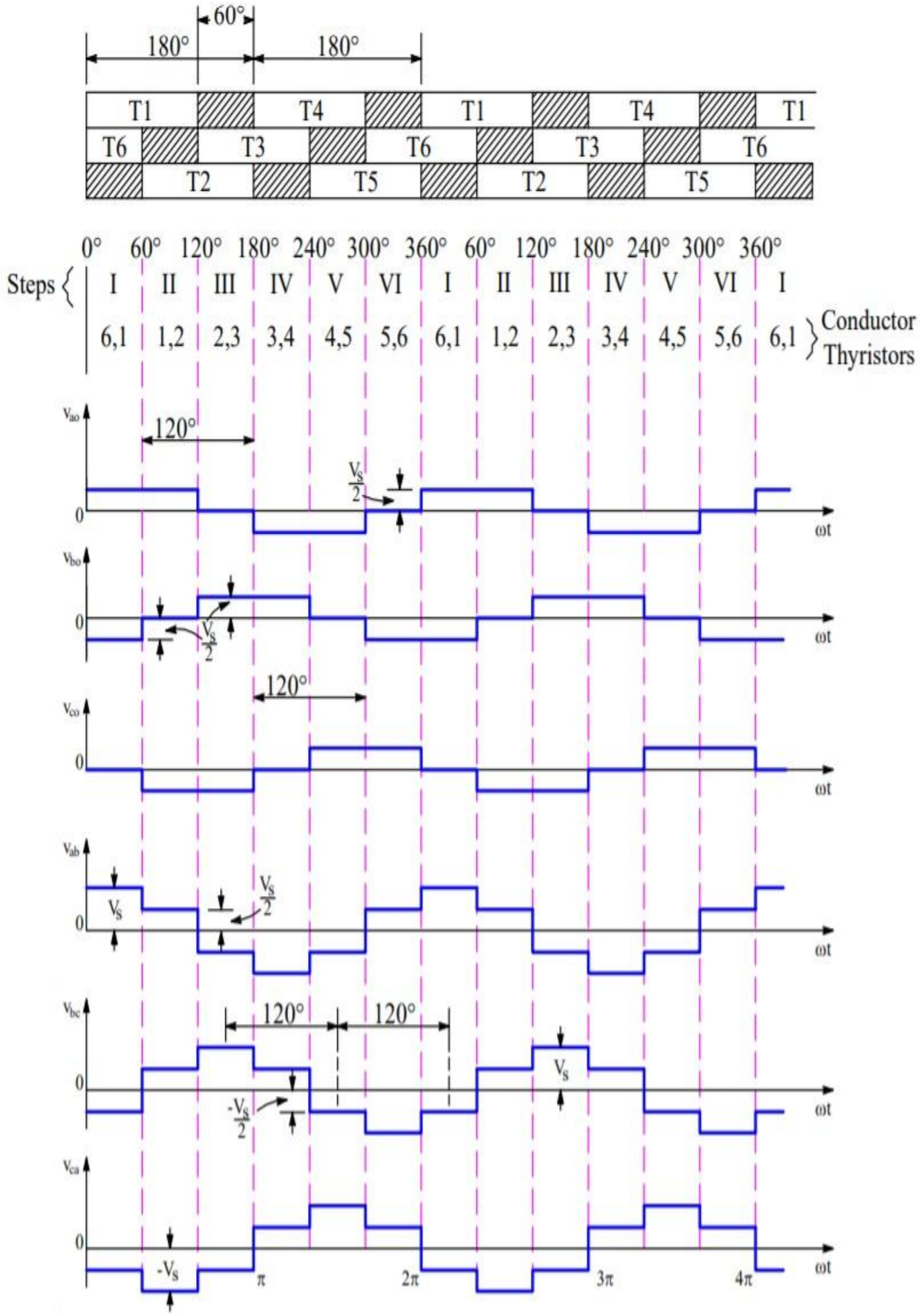
    Set Frequency HZ

16. Press SW3 for frequency increment & SW4 for frequency decrement - (Note the frequency varied from-10 to 50hz).
17. Switch "ON" MCB in IGBT Power Module.
18. Check the pulse output Test point connectors in IGBT power Module front Panel using CRO.
19. Ensure the availability of all 6 pulses using CRO.
20. Press SW3 switch to increase the FREQUENCY.
21. For every frequency note down output voltage across load & tabulate.
22. To reset the function, Press RST reset switch.

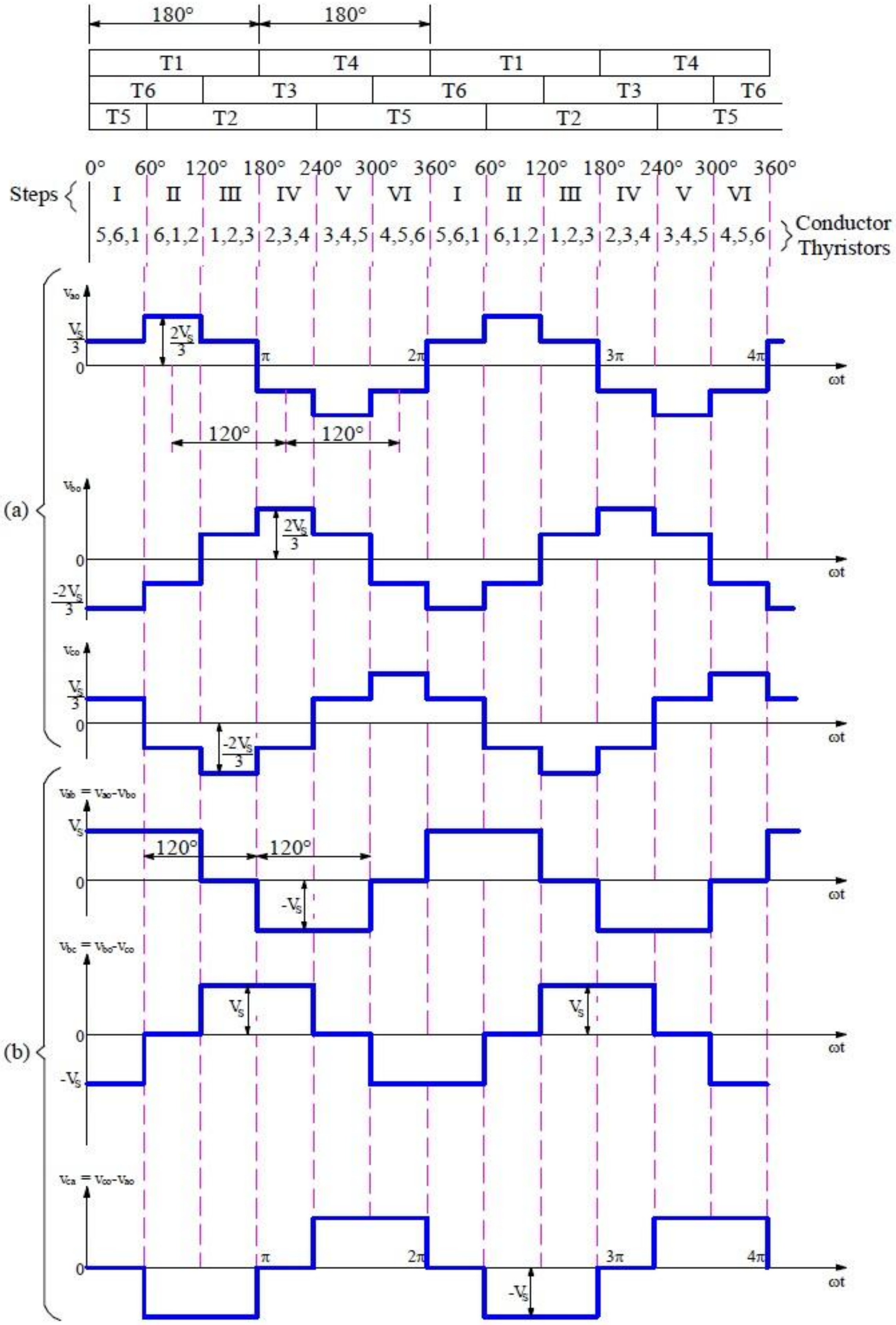
Tabular Column:

S.No.	Output Voltage (Volts)	Time (msec)	Modulation Index

Waveforms for 120^0 mode operation:



Waveforms for 180^0 mode operation:



Result:

Experiment: **Date:**

Study The Performance of Single-Phase Dual Converter

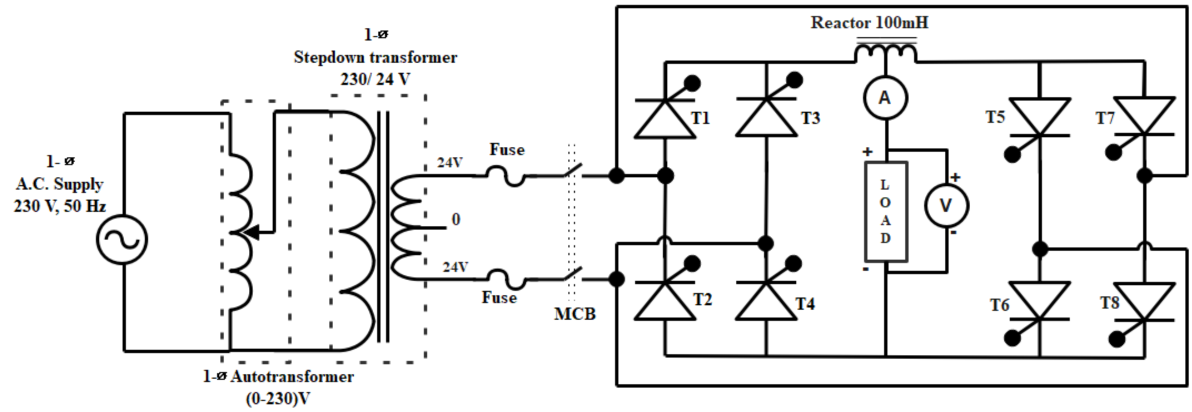
Aim:

To study the single phase fully controlled full bridge converter with R & RL load

Apparatus required:

S. No	Components	Specifications	Quantity
1	Single phase Dual power module		1 No
2	CRO, CRO probes		1 No
3	R, RL loads	100 Ω	1 No
4	Autotransformer	0-230 V	1 No
5	Step-down transformer	230 V / 48V	2 No
6	Microcontroller unit		1 No
7	Connecting wires		As per required

Circuit diagram:



Precautions:

1) Keep the input MCB is in off condition in SCR power module.
2) Keep the power ON/OFF switch is in off position in PLSC-01A.
3) Ensure the proper output connection of DC motor.
4) Connect 24V AC input supply to P_1, N_1 terminals (through isolation & auto transformer).
5) Connect 24V AC input supply to P_2, N_2 terminals (through isolation & auto transformer).
6) Check the PWM cable connection with controller & SCR power.
7) Connect the dspic programmer with micro controller.
8) Connect the DC motor field supply through field exciter box.

Procedure:

1) Connect single-phase input supply to SCR power module (back side).

2) Connect single-phase input supply to $P1_{IN}$, $N1_{IN}$, terminals in SCR power module through
Single-phase isolation transformer & autotransformer.
3) Connect single-Phase input supply to $P2_{IN}$, $N2_{IN}$, terminals in SCR power module through single-phase isolation transformer & autotransformer.
4) Connect G_1 (Pulse output) to G_1 (scr gate input) (all TWO to be shorted using patch chords
5) or 5A wires) Connect K_1 (Pulse output) to K_1 (ser gate input) (all TWO to be shorted using patch chords or 5A wires)-CONNECT ALL GI-G8 & K1-K8.
6) Connect IGR inductor [0%) terminal to connector [1] in front panel & [100 %] terminal to front panel connector [2].
7) Connect R load first terminal to IGR middle terminal (50%).
8) Connect R load second terminal to L_2 terminals in front panel SCR power module.
9) Connect 34 pin flat cable to microcontroller and SCR power module (pulse input connector).
10) Switch "ON" the power using power ON/OFF switch in microcontroller unit.
11) Switch "ON" the power using power ON/OFF switch in SCR unit.
12) Switch "ON" single-phase supply to SCR unit.
13) Check the ZCD square pulse in test point connectors in SCR power module front panel using CRO.
14) Switch on the power in microcontroller unit by using IRS switch.
15) in microcontroller unit the display will be
"1 Phase DUAL converter"
"Mode 1 (P)-forward
Mode 2 (N)-reverse"
16) Press SW3 switch to select "Mode-1" control.
17) The display will be

	"1. Open loop
	Press SW3 switch to select "open loop" control.
Firing angle	180 degree
Actual speed	0000 RPM

18) Now the Pulse is in minimum position.
19) Ensure the availability of all 4 pulses using CRO.
20) Switch ON the front panel MCB in SCR module and before switch ON keep the autotransformer in minimum condition, check the field supply connection of de motor.
21) Now slowly increase the autotransformer up to 80%
22) Press SW3 switch to increase the firing angle from 180-0 degree corresponding actual motor speed is displayed.
23) Tabulate the firing angle and actual output voltage in table-1 to reset the function, press SW5 reset switch.

24) Repeat the above procedure with Mode 2 control (reverse polarity) by pressing sw4 switch. note down the corresponding reading.

Tabular form:

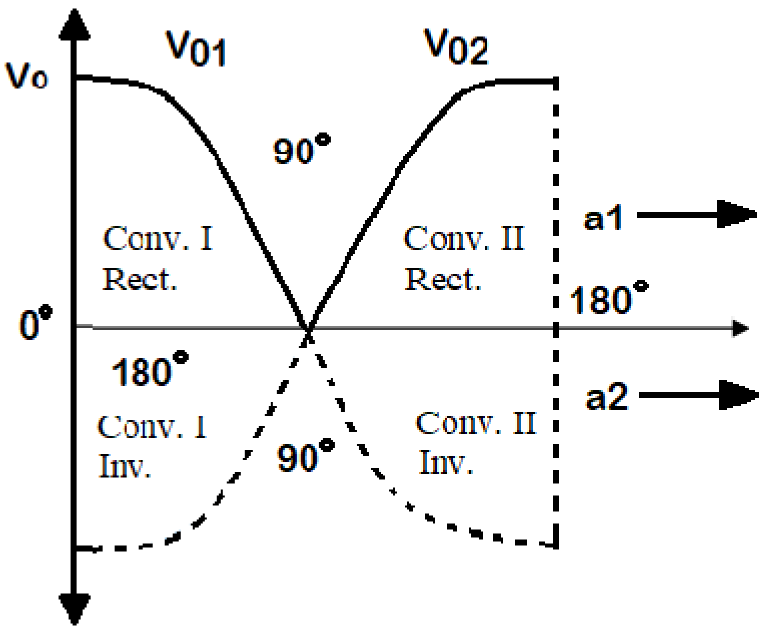
Mode-1

S. no.	Firing angle in degree	SCR converter output voltage
1		
2		
3		
4		

Mode-2

S. no.	Firing angle in degree	SCR converter output voltage
1		
2		
3		
4		

Waveforms:



Result: